Four-Dimensional Quasielastic Neutron Scattering (4D-QENS) Analysis of Correlated Ionic Conduction in $SrCl_2$

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Methods of elucidating the mechanisms of fast-ion conduction in solid-state materials are pivotal for advancements in energy technologies such as batteries, fuel cells, sensors, and supercapacitors. In this study, we examine the ionic conduction pathways in single crystal $SrCl_2$, which is a fast-ion conductor above 900 K, using four-dimensional quasielastic neutron scattering (4D-QENS). We explore both coherent and incoherent neutron scattering at temperatures above the transition temperature into the superionic phase to explore the correlated motion of hopping anions. Refinements of the incoherent QENS yield residence times and jump probabilities between lattice sites in good agreement with previous studies, confirming that ionic hopping along nearest-neighbor directions is the most probable conduction pathway. However, the coherent QENS reveals evidence of de Gennes narrowing, indicating the importance of ionic correlations in the conduction mechanism. This highlights the need for improvements both in the theory of ionic transport in fluorite compounds and the modeling of coherent 4D-QENS in single crystals.

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I. INTRODUCTION

Ionic conductivity plays a critical role in a wide range of energy applications, including batteries, fuel cells, sensors, and supercapacitors [1]. Fast-ion conductors, which exhibit high ionic mobility, are particularly valuable for enabling efficient charge transport in these applications [2,3]. Understanding the mechanisms of ionic conduction, including the interplay between structural dynamics, correlated motion, and defect-mediated pathways, is therefore pivotal for the design of next-generation materials [2–4]. Exploring these mechanisms is essential for advancing the design of safer, more efficient, and scalable energy-storage systems [2,3,5,6].

In many solid-state ionic conductors, mobile ions undergo random hopping processes, moving between vacant sites created either by doping or thermally activated Frenkel defects [6,7]. However, there is evidence in some materials of the importance of metastable interstitial sites along the hopping pathway [2,8,9] or cooperative ionic motion, where the movement of one ion facilitates the motion of nearby ions by lowering the energy barrier for subsequent hops [2,10]. While simulations have highlighted the significance of ionic correlations in conduction mechanisms, experimental techniques capable of directly detecting and characterizing this phenomenon remain underdeveloped [2,11–14].

One of the primary microscopic techniques for probing ionic dynamics in solid-state conductors is quasielastic neutron scattering (QENS) [15–17]. QENS is most frequently used to study the hopping processes of mobile ions in polycrystalline materials and analyzed using jump diffusion models, such as the Chudley-Elliott model [18], to yield spherically averaged values of hopping rates. There are fewer single crystal studies, and these have been confined to measurements along high-symmetry directions due to instrumental limitations [6,7]. However, the discovery of ionic conductors with more complex hopping pathways [2] and correlated ionic motion [19,20], require these limitations to be overcome in order to advance our understanding of ionic diffusion mechanisms in next-generation conductors.

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We propose that 4D-QENS offers a promising avenue for addressing this gap. Over the past decade, it has become increasingly common to perform time-of-flight inelastic neutron scattering with continuous or quasi-continuous sample rotation to generate four-dimensional hypervolumes of scattering in reciprocal space, **Q**, and energy transfer, ω [21]. This allows the energy dependence of scattering at constant-**Q** to be extracted, unlike conventional time-of-flight spectroscopy measured at a fixed sample angle. So far, these measurements have been used to generate 4D-S(**Q**, ω) for the analysis of excitation spectra, providing methods of exploring, for example, phonon anharmonicity or intraband transitions in correlated electron systems [22,23]. However, in this paper, we show that it is just as valuable in the analysis of QENS measurements, allowing the **Q** dependence of quasielastic linewidths to be determined over substantial reciprocal space volumes, not just high symmetry directions, enabling more stringent comparisons with theoretical models.

While conventional neutron-scattering methods have been effective in analyzing incoherent QENS in terms of jump diffusion models [7], measurements of coherent QENS have so far provided much more limited information. Unlike incoherent scattering, which probes the hopping of individual ions, coherent scattering can reveal the impact of ionic correlations on the hopping process. The coherent QENS intensity, $S_{coh}(\mathbf{Q})$, measures two-particle correlations, which are sensitive to local relaxations around both the hopping ions and the vacancies they may leave behind. On the other hand, coherent QENS linewidths, $\Gamma_{coh}(\mathbf{Q})$, are sensitive to the lifetimes of these correlations. If the local relaxations are much faster than the residence times, then it may be valid to analyze the coherent linewidths in terms of jump diffusion models in the same way as incoherent scattering [24]. However, the linewidths may also reveal a much more complex interplay of time scales, especially when there are metastable sites along the hopping pathway. It is well known from studies of liquids and amorphous materials that there can be a narrowing of $\Gamma_{coh}(\mathbf{Q})$ close to peaks in $S_{coh}(\mathbf{Q})$, a feature known as de Gennes narrowing [25], which reflects the increased stability of more probable ionic correlations. There are far fewer observations of de Gennes narrowing in crystalline materials, although it has been predicted to apply to a model of dilute interstitials [26]. In this paper, we propose that that the observation of de Gennes narrowing in ionic conductors provides evidence of metastable ionic configurations in the ionic hopping process.

This study uses 4D-QENS measurements to investigate ionic conductivity in the superionic phase of a fluorite compound, $SrCl_2$, which has been investigated using more conventional neutron scattering methods in the past [6,7]. Our results confirm the applicability of the Chudley-Elliott jump diffusion model to describe the incoherent scattering of hopping anions [7,18], which dominates the scattering at low **Q**, with improved accuracy because of the continuous three-dimensional coverage in reciprocal space. However, this work provides fundamentally new insights into the role of ionic correlations revealed by the **Q**-dependent coherent linewidths, in particular the observation of de Gennes narrowing, providing evidence of metastable ionic configurations during the hopping process that were not predicted by previous theoretical studies [14,27]. This demonstrates the power of 4D-QENS in revealing details of ionic conduction mechanisms that are impossible to determine by other experimental techniques.

II. EXPERIMENT

$SrCl_2$ was intensively studied by the Hutchings group over 40 years ago, along with a number of other fluorite compounds that exhibit fast-ion conduction just below their melting temperature [7]. $SrCl_2$ is ideal for investigating ionic conductivity because strontium is a nearly perfect coherent scatterer ($\sigma_{coh} = 6.19$ barns, $\sigma_{incoh} = 0.06$ barns), whereas chlorine has strong coherent and incoherent cross sections ($\sigma_{coh} = 11.53$ barns, $\sigma_{incoh} = 5.3$ barns). In the fast-ion phase, the strontium ions are considered to remain on the regular cation sites, so both the coherent and incoherent scattering away from the Bragg peaks are dominated by the mobile chlorine anions, whose diffusion generate the quasielastic energy broadening.

The $SrCl_2$ crystals used in our experiments were grown using a commercial reagent without additional purification. Ten grams were vacuum-sealed in a carbon-coated silica tube and crystal growth was carried out by the electrodynamic gradient method using a 25-zone furnace. The section of the resulting crystal used in the experiment was clean and transparent. Further details are given in the Supplemental Material [28].

Measurements of the diffuse scattering in $SrCl_2$ were performed on the CORELLI instrument at the Spallation Neutron Source [29]. This allows the energy-integrated diffuse scattering, S(**Q**) to be determined over a large reciprocal space volume as a function of temperature. The sample, which was $15 \times 5 \times 5$ mm^3 in size, was mounted in a high-temperature furnace for measurements between 800 and 1100 K. In agreement with earlier studies [7], the most significant diffuse scattering is seen beyond the (200) Bragg peak ($|Q| \approx 1.75$ Å^{-1}), with an intensity that grows strongly on entering the superionic phase (Fig. 1).

The QENS measurements were performed on the Cold Neutron Chopper Spectrometer (CNCS) at the Spallation Neutron Source [30]. The sample was mounted in a furnace on a rotating stage that allowed scans over 360° range in steps of 1°. This rotation allowed the reconstruction of a full 4D volume of data covering **Q** space and energy transfer. Both the CORELLI and CNCS data were reduced and symmetrized using the Mantid software suite [31,32].

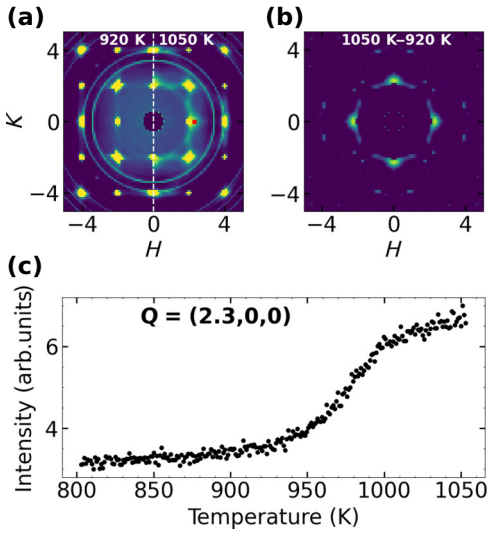


FIG. 1. (a) Energy integrated neutron diffraction map in the [HK0] scattering plane of $SrCl_2$ measured on CORELLI at 920 K (left) and 1050 K (right). (b) The difference between the scattering intensity at 1050 K and 920 K. (c) The temperature dependence of the diffuse scattering intensity at $\mathbf{Q} = (2.3, 0, 0)$, indicated by the red square.

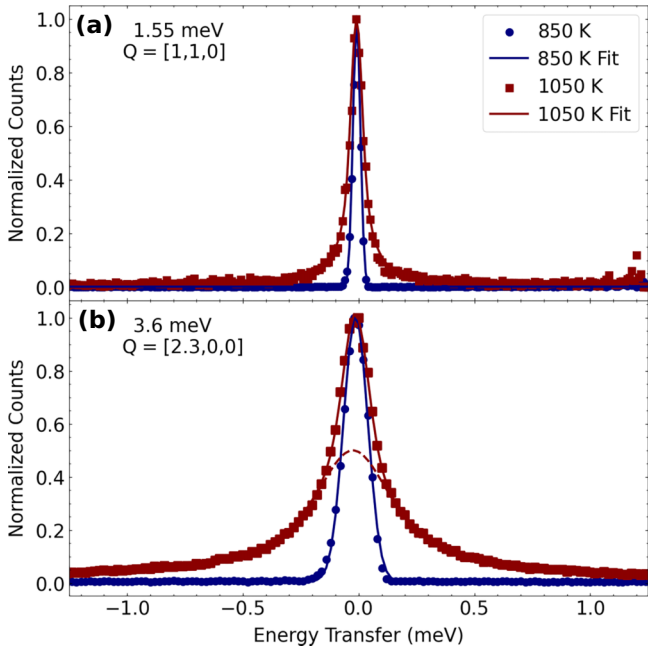


FIG. 2. (a) Energy distributions for incident energy 1.55 meV measured at the $\mathbf{Q} = [1, 1, 0]$ position at 1050 K, above the ionic conduction transition, and at 850 K, below the ionic transition. This can be compared with (b) 3.6-meV incident energy measurements of $\mathbf{Q} = [2.3, 0, 0]$, the region highlighted by a red box in Fig. 1(a), showing larger energy broadening due to the coherent scattering represented by a dashed line.

For reasons to be discussed in the next section, we performed the CNCS measurements with two different incident energies. By using an incident energy of 1.55 meV on CNCS, we were able to fit QENS up to $|Q| \approx 1.61$ Å^{-1}, with an energy resolution of 15.4 μeV, determined from scans measured well below the superionic transition temperature at 850 K [Fig. 2(a)]. With an incident energy of 3.6 meV, we extended the Q coverage of our fits up to $|Q| \approx 2.18$ Å^{-1}, with an energy resolution of 49.8 μeV [Fig. 2(b)].

III. RESULTS

The measurements performed on CORELLI confirm that there is a growth in coherent diffuse scattering associated with the superionic phase above approximately 920 K (Fig. 1). The 4D-QENS measurements on CNCS were therefore taken at 1050 K, about 100 K below the melting temperature. The 4D grid of scattering intensity following data reduction using the Mantid package was divided into cubes of 0.1 reciprocal lattice units (r.l.u.) in order to model the quasielastic broadening at each $\mathbf{Q}$. $S(\mathbf{Q}, \omega)$ consists of contributions from elastic, quasielastic, and inelastic components, as well as instrumental background [15,16,33]:

$$\mathbf{S}(\mathbf{Q},\omega) = [A_0(\mathbf{Q})\delta(\omega) + A_1(\mathbf{Q})S_{\text{incoh}}(\mathbf{Q},\omega) + A_2(\mathbf{Q})S_{\text{coh}}(\mathbf{Q},\omega) + B(\mathbf{Q},\omega)] \circledast R(\mathbf{Q},\omega). \quad (1)$$

In this equation, $A_0(\mathbf{Q})$ is the intensity of purely elastic scattering mostly from instrumental backgrounds as well as very weak strontium incoherent scattering. $A_1(\mathbf{Q})$ and $A_2(\mathbf{Q})$ are the incoherent and coherent scattering contributions, respectively, from the chlorine ions. Both are broadened in energy by the chlorine ion diffusion, modeled by Lorentzian lineshapes of differing half-widths, $\Gamma_{\text{incoh}}(\mathbf{Q})$ and $\Gamma_{\text{coh}}(\mathbf{Q})$, respectively. The final contribution, $B(\mathbf{Q}, \omega)$, represents inelastic contributions, mostly from phonon and multiphonon scattering, but in practice, it was too small at such low energies and wave vectors to be included in our analysis apart from a constant background. All components of the scattering are broadened by a Gaussian resolution function, $R(\mathbf{Q}, \omega)$, whose standard deviations at each incident energy were determined by measurements at 850 K. Quasielastic lineshapes were therefore fit to Voigt functions, which result from the convolution of the Lorentzian and Gaussian components, with the Gaussian width fixed by the resolution function.

A. Incoherent scattering

When analyzing 4D-QENS, it is not possible *a priori* to separate the coherent and incoherent components in the scattering cross section. However, as discussed by Hutchings *et al.* [7], the energy broadening of incoherent scattering is predicted to be much smaller than coherent

scattering. This is because incoherent energy broadening results from the time dependence of the self correlation function, so it reflects the average residence times of all the anions. On the other hand, the energy broadening of coherent diffuse scattering results from the time dependence of two-particle correlation functions involving defects in the anion sublattice, either vacancies or interstitials or both, along with relaxations of neighboring ions, during the hopping process. If, for example, only 10% of the anion sites are involved in hopping at any one time, *i.e.*, they host a vacancy or interstitial anion, the energy scale of coherent QENS will be an order-of-magnitude larger than incoherent QENS (although both are **Q** dependent, so the ratio is not precise).

As shown in the next section, the coherent diffuse scattering is weak at low **Q**, below the (200) Bragg peak, so the high resolution measurements were dominated by incoherent scattering [28]. An example of scattering above and below the transition temperature measured with an incident energy of 1.55 meV can be seen in Fig. 2, showing a clear broadening at 1050 K, which is absent at 850 K. Using the least-squares method [34], we fit such data to the sum of a Voigt and a Gaussian function, with the Gaussian standard deviation of both functions constrained to be equal to the estimated resolution. By automating such fits over the measured volume in reciprocal space, we were able to generate a 3D map of $\Gamma_{\text{incoh}}(\mathbf{Q})$, without being confined to high-symmetry directions. The resulting Q dependence of the incoherent linewidth in the $[H, K, 0]$ plane is shown in Fig. 3.

These Q maps were then fit in three dimensions to the Chudley-Elliott (CE) jump diffusion model [18], which assumes that the ions jump instantaneously between lattice

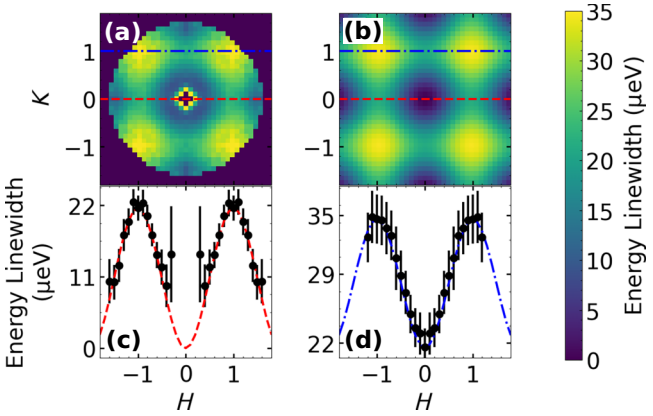


FIG. 3. Plots of the $L = 0$ plane of (a) the incoherent linewidths measured using incident energy $E_i = 1.55$ meV compared to (b) the Chudley-Elliott model with parameters $p_1 = 0.754 \pm 0.002$, $p_2 = 0.746 \pm 0.002$, and $\tau_{\text{inc}} = 25.36 \times 10^{-12}$. One-dimensional line cuts taken at (c) $K = 0$ and (d) $K = 1$ further illustrate the quality of fit between the measured linewidths (points) and the model (dashed line).

sites. The model predicts that the linewidths will increase quadratically at low Q, but then fall to zero at all the wave vectors corresponding to the hopping sublattice [35,36]:

$$\Gamma_{\text{incoh}}(\mathrm{Q}) = \frac{1}{2\pi\,\tau_{\text{incoh}}} \sum_i \frac{p_i}{n_i} \left(\sum_j [1 - \exp(-i\mathbf{Q} \cdot \mathbf{r}_{ij})] \right). \tag{2}$$

In this equation, τ_{incoh} is the mean residence time of anions between hops, p_i is the probability of hopping to the ith nearest neighbor, n_i is the number of equivalent ith nearest neighbors, and r_{ij} is the interatomic vector connecting sites i and j. In $SrCl_2$, $i = 1$ and 2 correspond to hops between nearest neighbor anion sites ($r_{ij} = a_0/2 \approx 3.6$ Å, $n_i = 6$) and next-nearest neighbor anion sites ($r_{ij} = a_0/\sqrt{2} \approx 5.1$ Å, $n_i = 12$), respectively.

As shown in Fig. 3, our data strongly supports the CE model predictions, with four lobes centered symmetrically around $\mathrm{Q} = 0$. The results of this analysis are compared to the results in (a) using 1D line cuts along the $[H, 0, 0]$ and $[H, 1, 0]$ directions, as presented in (c) and (d) respectively. Based on our fits, we find that $p_1 = 0.754 \pm 0.002$, $p_2 = 0.246 \pm 0.002$, and $\tau_{\text{incoh}} = 25.36 \times 10^{-12}$ s. These values are consistent with previous experimental estimates in $SrCl_2$ [7,35,36].

We attempted to include additional hopping vectors, but the values of p_n were 0 within errors for any $n \geq 3$. We also checked if the anions might hop to and from "interstitial" sites, such as those predicted by the Hutchings group [7], with similar results. It is worth noting that such additional hops can substantially distort the shape of the four lobes shown in Fig. 3(a), so we believe the conclusion that only nearest and next-nearest neighbor hops have a significant probability is more robust in the 4D-QENS analysis than in previous measurements that were confined to a few high-symmetry directions [35].

B. Coherent scattering

We used the results of the CE model refinement to constrain our analysis of the 3.6 meV data at 1050 K, in which there are significant contributions from coherent scattering. With the extended energy range, we fit the data to the sum of a Gaussian function, from purely elastic scattering, and two Voigt functions, representing the incoherent and coherent quasielastic scattering, respectively. However, the parameters of the incoherent Voigt function linewidths were fixed to those predicted by the CE model, allowing the coherent contributions to be determined without excessive parameter correlations. The fit results are shown in Fig. 4, with (a) showing the incoherent structure factor, (b) the incoherent linewidths, (c) the coherent structure factor, and (d) the coherent linewidths. Note that the incoherent linewidths shown in Fig. 4(b) are calculations based on

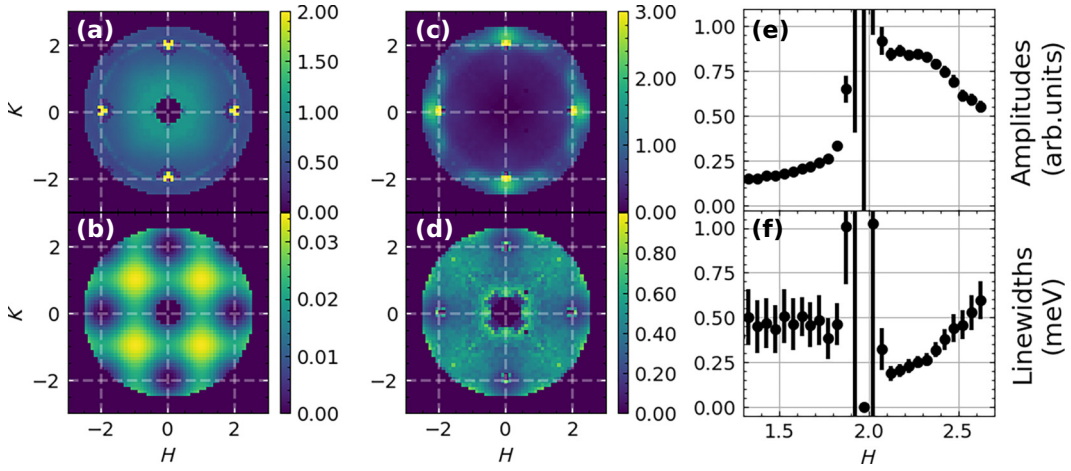


FIG. 4. Results of fits to the incoherent and coherent scattering measured with $E_i = 3.6$ meV. Shown are the $L = 0$ planes of (a) the incoherent structure factor, (b) the incoherent linewidths in meV generated using the Chudley-Elliott model parameters shown in Fig. 3, (c) the coherent structure factor, (d) the coherent linewidths in meV. Additional fits were performed with reduced bin sizes along the H direction, with the coherent amplitudes of these fits shown in (e) and the linewidths shown in (f).

the CE model derived from the 1.55-meV data, whereas Fig. 4(a) are fits, whose Q dependence is consistent with a monotonic fall in the integrated incoherent intensity from the Debye-Waller factor.

Figure 4(c) shows the integrated coherent intensity, which is consistent with the CORELLI data shown in Fig. 1(b), as well as previous neutron measurements [7]. However, the most interesting result is shown in Fig. 4(d). Although the coherent intensity is weak at wave vectors below the (200) Bragg peaks, it is nevertheless reliably determined from the fits, allowing the coherent linewidths to be examined over the entire volume in reciprocal space for the first time. It is clear that the $\Gamma_{\rm coh}$ bears no resemblance to $\Gamma_{\rm incoh}$, which falls to zero symmetrically around $\mathbf{Q} = (200)$. Instead, there is a strong asymmetry, with a significant dip in the linewidths at wave vectors beyond the (200) Bragg peaks. Line cuts along the longitudinal direction in Q show a broad maximum in coherent diffuse intensity at $H \gtrsim 2.2$ [Fig. 4(e)], correlated with a comparable reduction in the linewidths [Fig. 4(f)], although the minimum wave vector is not as clearly defined in our analysis. This is characteristic of de Gennes narrowing [25], in which the linewidths are minimized at wave vectors corresponding to maxima in the diffuse $S_{\rm coh}(\mathbf{Q})$. This will be discussed in the next section.

IV. DISCUSSION

Examples of de Gennes narrowing have mostly been confined to liquids, including molten $SrCl_2$ [37], and glasses [38], but the only microscopic derivation in crystalline materials has been provided by Sinha and Ross [26], who modeled the solid-state diffusion of a dilute concentration of hydrogen interstitials. Using linear response theory, they predicted that the coherent linewidths could be derived from the incoherent linewidths by the following equation [16,26]:

$$\Gamma_{\rm coh}(\mathbf{Q}) = \frac{c(1-c)\Gamma_{\rm incoh}(\mathbf{Q})}{S_{\rm coh}(\mathbf{Q})} \tag{3}$$

where c is the concentration of interstitials. However, Eq. (3) is inconsistent with the global $\mathbf{Q}$ dependence shown in Figs. 4(b)–4(d). We suggest that this is due to the neglect of ionic correlations in the dilute limit [17]. Nevertheless the observed coincidence of a maximum in $S_{\rm coh}(\mathbf{Q})$ with a minimum in $\Gamma_{\rm coh}(\mathbf{Q})$ is consistent with the intuitive interpretation of de Gennes narrowing that more probable interionic correlations are associated with an increased lifetime of the corresponding ionic configurations.

The reason for the increased lifetimes goes to the heart of the ionic conduction mechanism. In the Hutchings study [7], the diffuse scattering was modeled in terms of anion clusters, comprising an interstitial anion, located half-way between the regular anion sites but shifted towards a neighboring empty cube center, as well as a number of neighboring anions relaxing away from it. Whether these clusters arose from thermally activated Frenkel defects that were not associated with vacancy hopping or were part of the hopping process itself could not be determined from their measurements. This cluster model was, however, challenged by molecular dynamics (MD) simulations based on realistic pair potentials [27,39], which were able to reproduce the measured $S(\mathbf{Q})$ from an average over the instantaneous ionic configurations, without seeing any evidence of the proposed interstitial clusters in the anion trajectories. Instead, Gillan proposed that anions randomly hopped directly from one regular site to another [11], producing an equal number of vacancies and doubly occupied sites (which Gillan called "interstitial" sites), where the

occupation number is defined by a large sphere centered on each site [39]. According to the MD simulations, the strong peaks in diffuse scattering beyond the (200) Bragg peaks are produced predominantly by correlations between anions on the doubly occupied sites, both of which are presumably strongly relaxed from the site center. On the other hand, relaxations of neighboring ions around the vacancy sites mostly contribute to $S_{coh}(\mathbf{Q})$ in regions where the scattering is relatively weak.

The strong **Q** dependence of the coherent linewidths supports the conclusion that the peak in $S_{coh}(\mathbf{Q})$ is associated with the hopping process itself, since thermal activation of noninteracting clusters would produce a **Q**-independent linewidth corresponding to their average lifetime. Furthermore, the observation of de Gennes narrowing implies that there is a finite probability for anions to be trapped on doubly occupied sites until one of them is able to hop to a neighboring vacancy site. The MD simulations show evidence of chains of correlated hops consistent with this scenario (e.g., see Fig. 11 of Ref. [40]), resulting from short-range interactions within a unit cell rather than the longer range correlations postulated in other materials [38]. With a dilute concentration of thermally activated vacancies, the residence time of doubly-occupied sites could be substantially longer than the residence time of vacancies determined from the coherent broadening below the (200) Bragg peaks [Fig. 4(f)].

V. CONCLUSION

Our study shows that 4D-QENS, a variation of QENS, in which single crystal samples are continuously rotated through 360° while collecting data as a function of neutron time-of-flight, is a viable method for characterizing ionic hopping mechanisms in fast-ion conductors such as $SrCl_2$. It allows the reconstruction of $S(\mathbf{Q}, \omega)$ in four dimensions to facilitate comparisons with models of both incoherent and coherent cross sections as a function of wave-vector transfer. The high-resolution measurements of incoherent linewidths presented here largely confirm the results of earlier investigations of the incoherent scattering, which were analyzed in terms of the Chudley-Elliott jump diffusion model [7]. However, the previous measurements were confined to high-symmetry directions in reciprocal space, which would not have been as sensitive to additional hopping terms, such as to hypothetical interstitial sites. Conclusions that the only significant hopping probabilities are between nearest-neighbor and next-nearest-neighbor anion sites are therefore more robust in the present study where off-symmetry directions are included.

Our measurements of the coherent diffuse scattering are also broadly consistent with the earlier studies, with the most significant intensity observed at wave vectors above the (200) Bragg peaks. However, 4D-QENS has elucidated the **Q** dependence of the linewidths in unprecedented detail, covering a broad dynamic range of diffuse scattering intensities in a fine **Q** mesh. This has provided evidence of de Gennes narrowing, resulting from the enhanced lifetimes of defects, in which two anions are centered on the same site. We hope that this work will stimulate an extension of the theory of de Gennes narrowing in crystalline materials to systems, in which there are strong interactions between the diffusing ions, as well as future experiments utilizing the 4D-QENS method to investigate ionic diffusion pathways in more complex ionic conductors.

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DATA AVAILABILITY

The data that support the findings of this article are not publicly available upon publication because it is not technically feasible and/or the cost of preparing, depositing, and hosting the data would be prohibitive within the terms of this research project. The data are available from the authors upon reasonable request.

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